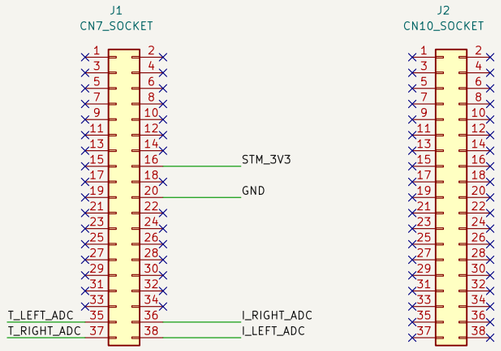
TMR MONITOR SHIELD | RevA.0 FUNCTIONAL LAYOUT

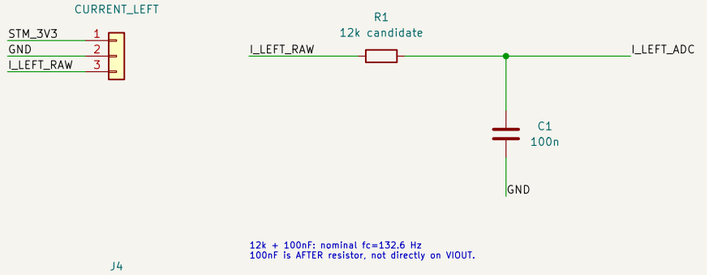
DRAFT – NOT FOR FABRICATION OR CONNECTION
Signal circuits only: current-sensor power path remains off-board.

01 NUCLEO-F446RE INTERFACE



CN7: 16=3V3, 20=GND
35=PC2, 37=PC3
36=PC1/PB9*, 38=PC0/PB8*
* Verify solder bridges; preserve IMU PB8/PB9.
CN10 pins intentionally unused by this shield.
NC marks do not alter the NUCLEO wiring.

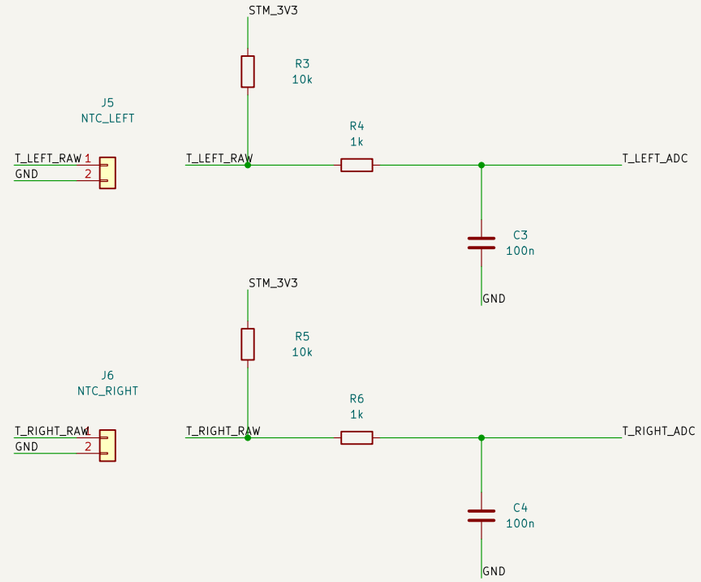
02 CURRENT INPUTS



12k + 100nF: nominal fc=132.6 Hz
100nF is AFTER resistor, not directly on VIOUT.

12k + 100nF: nominal fc=132.6 Hz
100nF is AFTER resistor, not directly on VIOUT.

03 NTC INPUTS



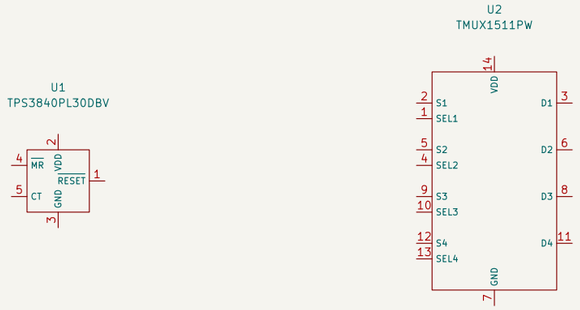
04 VOLTAGE INPUTS – RESERVED FOR NEXT STEP

BATTERY SIDE | J7 / R7–R11 / C5 – TO BE ADDED

MOTOR SUPPLY SIDE | J8 / R12–R16 / C6 – TO BE ADDED

Layout space only. Voltage dividers and their nets are not placed yet.

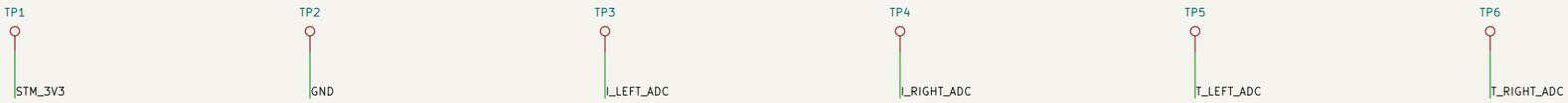
05 VOLTAGE SUPERVISOR / SWITCH



OPEN ELECTRICAL ITEMS
– Input ESD / fault protection and unpowered backfeed not designed yet.
– 3V3 supply margin, carrier decoupling and cable capacitance to be checked.
– ADC acquisition/leakage budget and sensor disconnection handling remain open.
– No automatic motor stop threshold is defined by this sheet.

OPEN MECHANICAL ITEMS
– Connector models and resistor/capacitor body envelopes are provisional.
– Template header positions only; actual socket height / clearance not verified.
– Keep RESET / USER / jumpers / ST-LINK access and cable strain relief.
– A clean ERC is not proof of these electrical or mechanical conditions.

06 TEST POINTS



NOT FOR FABRICATION – protection / mechanical review open

Sheet: /
File: TMR_Monitor_Shield_RevA.kicad_sch

Title: TMR Monitor Shield

Size: A3	Date: 2026-10-23	**Rev: RevA.0-DRAFT**
KiCad E.D.A. 10.0.5		Id: 1/1